Fixed-Edit FIBER–GRAND: Final Novelty, Theory, and Potential Gate

Exact membership-only aggregate inverse decoding, moment/tail laws, and a typical-versus-mean complexity phase diagram

Scientific de-risking dossier, version 3.0

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Authoritative scientific status

The original broad FIBER–GRAND proposal must remain narrowed. "Fiber guesswork" is ordinary posterior conditional guesswork; best-path versus best-string inference, hidden-path summation, priority-first search, and generic candidate-likelihood guessing are established. The surviving candidate is a narrower theorem-and-algorithm synthesis for a *fixed number of synchronization edits*: a codebook-free membership-only aggregate inverse generator, a strict unseen-candidate certificate, fixed-edit history moment and tail laws, and an operating-region comparison with code-side exact search.

The internal reference run of this package returns

`CONTINUE_HIGH_POTENTIAL_FIXED_EDIT_FIBER_TRACK_REAL_TRACE_REQUIRED`

This means *high potential, not field-defining status.* A measured post-front-end trace, independent external proof review, and independent novelty review remain mandatory. Any failure at that next gate is an immediate stop command.

Contents

1 Purpose and frozen decision discipline

The earlier computational gates established three facts. First, pathwise first-hit decoding can disagree with aggregate maximum likelihood (ML) on deletion channels. Second, a membership-first aggregate inverse decoder can be made exact through a strict residual-mass certificate. Third, at short lengths and low substitution probabilities, the decoder can be competitive with code-specific exact searches while using only syndrome or checksum membership rather than a materialized codebook.

Those results did not settle the most important field-defining questions:

(i) whether any genuinely new theorem survives the prior-art boundary;

(ii) whether typical, mean, and tail complexity have the same operating region;

(iii) whether the finite-length computational advantage is consistent with the new theory rather than a Python artifact;

(iv) whether a physically calibrated synchronization impairment occupies the favorable regime.

This dossier develops the first three questions rigorously and supplies a reproducible gate for the fourth. It does not permit the broad claim to be rescued by changing terminology after a negative result.

1.1 Three possible outcomes

CONTINUE

`CONTINUE_HIGH_POTENTIAL_FIXED_EDIT_FIBER_TRACK_REAL_TRACE_REQUIRED`. The narrow novelty, exactness, moment/tail theory, phase diagram, and strong-baseline signal all survive. Only paper-grade compiled implementation, external proof/novelty review, and one measured post-front-end trace are authorized.

NARROW

`NARROW_TO_FIXED_EDIT_THEOREM_AND_ALGORITHM_PAPER`. The mathematics survives but the scalable decoder advantage does not. A bounded-edit theorem/algorithm paper may proceed; the multi-year field-defining investment stops.

STOP

`STOP_FIBER_FIELD_DEFINING_PROGRAMME`. A novelty, correctness, or moment/tail theorem fails, or strong exact baselines systematically dominate in the theoretically favorable regime. No hardware, general-transducer, or broad systems programme is permitted.

2 Narrowed novelty adjudication

2.1 Claims withdrawn permanently

The following are not new contributions of the revised programme.

1. **Posterior candidate rank.** Ranking x by $W(y \mid x)$ under a uniform input prior is ordinary conditional guesswork. Its Rényi and large-deviation theory has established foundations [4, 5].

2. **Best path versus best string.** Ambiguous probabilistic automata distinguish a most-probable path from a most-probable emitted string, and the general most-probable-string problem is NP-hard for probabilistic automata [11].

3. **Priority-first or sequential search.** Priority-first ML decoding of linear block codes and synchronization-error sequential decoding are classical or established lines [6, 7, 8].

4. **Summing hidden alignments.** Forward-backward and weighted-transducer methods sum alignment paths, including in insertion/deletion/substitution receivers [9, 10].

5. **Generic candidate-likelihood guessing.** Guessing-based decoders for general discrete memoryless channels can rank input candidates by channel closeness and query code membership [2]; modern work also unifies noise-side and codeword-side guessing [3].

6. **Unrestricted transducer tractability.** General ambiguous automata can be computationally hard or undecidable, while bounded ambiguity supports only qualified algorithmic results [11, 12].

2.2 Closest insertion/deletion prior art

The narrow audit also found substantial overlap in synchronization coding:

- Gallager's 1961 report and later drift-state sequential decoders already search synchronization hypotheses [6, 8].
- A 1990 patent enumerates insertion/deletion "error frames," decodes each frame, computes a word likelihood, and selects a maximum-probability output [13].
- Fixed-k deletion work derives multiplicity-aware ML/ML* rules for one and two deletions [14].
- GC/GC+ codes and VT codes provide specialized polynomial or linear-time decoders for fixed deletions [15, 16].
- Polar guard-band constructions achieve information rates on insertion/deletion/substitution channels in an asymptotic, code-designed regime [17].
- GRAND-assisted modulation already applies lightweight GRAND ideas to a specialized variable-length demapping problem [18].

2.3 Residual novelty classification

No located primary source contains the complete narrow package below as one theorem-and-algorithm result:

> For a fixed number of uniformly located deletions followed by survivor BSC substitutions, generate inverse histories in probability order; canonicalize candidates; query only a code-membership oracle; exactly score only discovered codewords; terminate by a strict sum-of-stream-frontiers certificate; and characterize the history-revelation moments, tails, and rate-dependent operating region.

The appropriate classification is therefore

$$\boxed{\texttt{NARROW\_NOVELTY\_SURVIVES},}$$

not "a wholly new decoding principle." The shell and moment theorems may be viewed as elegant but elementary unless joined by a meaningful operating-region theorem and convincing implementation evidence. No search can exclude unpublished work or every patent family; an external novelty review remains required.

Table 1: Claim-by-claim novelty boundary.

Statement	Closest established line	Classification
Additive disturbance first-hit GRAND	GRAND [1]	Known
General channel candidate guessing	Guessing with abandonment [2]	Known
Best path versus aggregate string	Probabilistic automata [11]	Known
Priority-first exact search	Han et al. [7]	Known
Synchronization sequential decoding	Gallager; Banerjee et al. [6, 8]	Known
Enumerate edit frames, decode, score, select	US4922494A [13]	Partially anticipated
Fixed-k deletion multiplicity-aware ML	Sabary et al. [14]	Known ingredients
Membership-only aggregate inverse generation with strict sum-of-frontiers certificate	No complete primary result located	Distinct, narrow
Fixed-edit revelation moment/tail theorem	Guesswork supplies calculus, but no identical fixed-edit statement located	Distinct, narrow
Typical-versus-mean rate-complexity phase diagram	GRAND/GCD regime work is prior; this fixed-edit synthesis not located	Distinct synthesis

3 Channel, history representation, and exact decoder

3.1 Fixed-edit deletion/BSC channel

Let $x \in \{0,1\}^n$ and fix an edit count $t < n$. A deletion set $J \subseteq [n]$, $|J| = t$, is selected uniformly from

$$\mathcal{J}_t = \{J \subseteq [n] : |J| = t\}, \qquad S_n = |\mathcal{J}_t| = \binom{n}{t}.$$

The $m = n - t$ surviving symbols pass through a BSC with crossover probability $0 \le p < 1/2$, producing $y \in \{0,1\}^m$. For candidate x, let $d_J(x, y)$ be the Hamming distance between y and the subsequence of x obtained by deleting J. The exact likelihood is

$$\Lambda_y(x) = W(y \mid x) = \frac{1}{S_n} \sum_{J \in \mathcal{J}_t} p^{d_J(x,y)} (1-p)^{m - d_J(x,y)}. \tag{1}$$

For $p = 0$, each term is interpreted as the indicator of an exact subsequence match.

3.2 Inverse histories

An inverse history is

$$h = (J, b, e) \in \mathcal{H}_n := \mathcal{J}_t \times \{0,1\}^t \times \{0,1\}^m,$$

where b supplies the deleted symbols and e is a survivor substitution pattern. It generates

$$x_h = \mathrm{Ins}_J(y \oplus e, b)$$

with component weight

$$\omega(h) = \frac{1}{S_n} p^{\mathrm{wt}(e)} (1-p)^{m - \mathrm{wt}(e)}. \tag{2}$$

Algorithm 1 Certified fixed-edit FIBER search

Require: y, t, p, membership oracle for $\mathcal{C}$, history budget
initialize one probability-ordered stream for each $J \in \mathcal{J}_t$
$D \leftarrow \emptyset$; $s^\star \leftarrow -\infty$; $T^\star \leftarrow \emptyset$
while the global heap is nonempty **do**
  pop the largest next history; advance its stream
  construct x_h
  if $x_h \notin D$ **then**
    $D \leftarrow D \cup \{x_h\}$; query `IsCodeword`$(x_h)$
    if $x_h \in \mathcal{C}$ **then**
      compute exact $\Lambda_y(x_h)$ and update $(s^\star, T^\star)$
    end if
  end if
  $U_{\text{new}} \leftarrow \sum_{J \in \mathcal{J}_t} \eta_J$, where η_J is stream J's next weight
  if $T^\star \neq \emptyset$ and $s^\star > U_{\text{new}}$ **then**
    return the complete ML tie set $T^\star$ and the strict certificate
  end if
end while

For each candidate x, exactly one (b, e) is compatible with each J, hence

$$\Lambda_y(x) = \sum_{h: x_h = x} \omega(h).$$

The 2^t hidden assignments are inverse candidates, not extra channel-probability factors; this is why the sum of history weights over all inverse histories exceeds one, while each candidate likelihood remains correct.

3.3 Membership-first aggregate inverse decoder

For every deletion set J, create a stream that enumerates (e, b) in increasing substitution weight. Merge the stream heads in a max heap. Maintain a set of distinct candidates. At first discovery of x:

(a) query `IsCodeword`(x);

(b) only if true, compute the complete likelihood (1);

(c) update the incumbent ML score and discovered tie set.

Scoring discovered noncodewords is unnecessary: they cannot compete in the code-constrained maximization, and every undiscovered codeword remains covered by the global frontier bound.

Theorem 1 (Strict unseen-candidate certificate)**.** *Let η_J be the weight of the next unprocessed history in stream J, or zero if the stream is exhausted. Every completely unseen candidate x satisfies*

$$\Lambda_y(x) \le U_{\text{new}} := \sum_{J \in \mathcal{J}_t} \eta_J.$$

Suppose every discovered codeword has been exactly scored and let

$$s^\star = \max_{c \in \mathcal{C} \cap D} \Lambda_y(c), \qquad T^\star = \{c \in \mathcal{C} \cap D : \Lambda_y(c) = s^\star\}.$$

If $s^\star > U_{\text{new}}$, then

$$T^\star = \arg\max_{c \in \mathcal{C}} \Lambda_y(c).$$

Proof. For a fixed J, an unseen candidate has at most one compatible inverse history. Because that history has not been emitted, its weight is at most η_J. Summing over J gives the unseen-candidate bound. The strict inequality excludes every unseen codeword from tying or exceeding the incumbent. Discovered codewords have exact scores, so the discovered maximizers are precisely the global ML tie set. □

Remark 1 (Numerical certification). The production implementation uses outward rounding with `nextafter` and a scale-aware margin before accepting the strict inequality. Exhaustive micro-world tests remain the authoritative correctness check. A paper-grade implementation should use interval or fixed-point outward-rounded likelihoods near the stopping boundary.

4 Deterministic fixed-edit shell certificate

Let E be the actual number of substitutions in the transmitted history and define

$$a = \frac{1-p}{p}, \qquad \ell_{n,t} = \left\lfloor \log_a \binom{n}{t} \right\rfloor, \qquad w_E = \min\{m, E + \ell_{n,t}\}.$$

Theorem 2 (Deterministic shell-work bound)**.** *For $0 < p < 1/2$, after all inverse histories with substitution weight at most w_E have been processed, the strict certificate in theorem 1 must have fired. Hence*

$$Q_{\text{FIBER}} \le 2^t \binom{n}{t} \sum_{i=0}^{w_E} \binom{m}{i}. \tag{3}$$

For $p = 0$,

$$Q_{\text{FIBER}} \le 2^t \binom{n}{t}.$$

Proof. The transmitted codeword is discovered no later than its actual deletion set, deleted bits, and survivor error mask. Thus the incumbent score is at least

$$\frac{1}{S_n} p^E (1-p)^{m-E}.$$

After every history through shell w has been emitted, the next history in each stream has weight at most

$$\frac{1}{S_n} p^{w+1} (1-p)^{m-w-1}.$$

Summing over the S_n streams gives

$$U_{\text{new}} \le p^{w+1} (1-p)^{m-w-1}.$$

At $w = E + \ell_{n,t} < m$, the ratio of this bound to the one-history incumbent lower bound is at most

$$S_n \left(\frac{p}{1-p} \right)^{\ell_{n,t}+1} < 1,$$

because $\ell_{n,t} + 1 > \log_a S_n$. If $w_E = m$, all streams are exhausted. Each stream contains $2^t \binom{m}{i}$ histories in shell i, yielding (3). At $p = 0$, only shell zero has positive weight. □

Corollary 1 (Fixed-edit typical history exponent)**.** *For fixed t and $p \in (0, 1/2)$, for every $\varepsilon > 0$,*

$$\Pr\left\{ \frac{1}{n} \log_2 Q_{\text{FIBER}} \le h_2(p) + \varepsilon \right\} \longrightarrow 1.$$

Proof. For fixed t, $\log \binom{n}{t} = O(\log n)$ and $\ell_{n,t} = O(\log n)$. The binomial error weight satisfies $E/m \to p$ in probability. Standard Hamming-ball estimates give

$$\sum_{i=0}^{qm} \binom{m}{i} \le 2^{m h_2(q)}, \qquad q < 1/2,$$

and continuity of h_2 completes the proof. □

5 History revelation: exact moment and tail laws

The previous result is a high-probability upper bound. It does not settle the mean or rare-tail complexity. To expose those quantities, define the rank at which the *actual inverse history* is revealed in the global probability ordering.

Definition 1 (History revelation rank). Let R_{hist} be the rank of the actual triple (J, b, e) among all inverse histories ordered by nonincreasing component probability, under an arbitrary deterministic tie order within each shell.

Define

$$K_n = 2^t\binom{n}{t}, \qquad B_m(k) = \sum_{i=0}^{\min\{k,m\}}\binom{m}{i},$$

with $B_m(-1) = 0$.

Proposition 1 (Finite-length rank sandwich). *Conditioned on $E = e$,*

$$1 + K_n B_m(e-1) \le R_{\mathrm{hist}} \le K_n B_m(e). \tag{4}$$

Proof. Every history in a lower-weight shell has strictly higher probability and must precede the actual history. At most all histories through shell e precede or include it. There are $K_n\binom{m}{i}$ histories in shell i. ☐

Theorem 3 (Fixed-edit moment exponent). *Fix t, $p \in (0, 1/2)$, and $\rho > 0$. Let*

$$\alpha = \frac{1}{1+\rho}.$$

Then

$$\lim_{n\to\infty}\frac{1}{n}\log_2 \mathbb{E}[R_{\mathrm{hist}}^{\rho}] = \rho H_\alpha(p), \tag{5}$$

where the binary Rényi entropy is

$$H_\alpha(p) = \frac{1}{1-\alpha}\log_2\left(p^\alpha + (1-p)^\alpha\right).$$

The certified decoder obeys the corresponding upper bound

$$\limsup_{n\to\infty}\frac{1}{n}\log_2 \mathbb{E}[Q_{\mathrm{FIBER}}^{\rho}] \le \rho H_\alpha(p). \tag{6}$$

Proof. The fixed-edit factor K_n is polynomial in n and therefore subexponential. By proposition 1, the exponential order of R_{hist} conditioned on $E \approx qm$ is $2^{mh_2(q)}$ for $q \le 1/2$; the maximizing type below is strictly less than $1/2$. The binomial method of types gives

$$\frac{1}{n}\log_2 \mathbb{E}[R_{\mathrm{hist}}^{\rho}] \longrightarrow \sup_{0\le q\le 1/2}\left\{\rho h_2(q) - D_2(q\|p)\right\},$$

where

$$D_2(q\|p) = q\log_2\frac{q}{p} + (1-q)\log_2\frac{1-q}{1-p}.$$

The optimizer is

$$q_\rho = \frac{p^{1/(1+\rho)}}{p^{1/(1+\rho)} + (1-p)^{1/(1+\rho)}},$$

and direct substitution yields the variational identity

$$\sup_q \{\rho h_2(q) - D_2(q\|p)\} = \rho H_{1/(1+\rho)}(p).$$

For the certified decoder, theorem 2 replaces E by $E + O(\log n)$ and multiplies by only a polynomial fixed-edit factor; this does not change the exponent. Since the decoder may stop earlier, only the upper bound is universal. □

Remark 2 (The crucial distinction). At $\rho = 1$, the mean history-revelation exponent is $H_{1/2}(p)$, not $h_2(p)$. A decoder may therefore have attractive median or fixed-quantile latency while suffering unfavorable mean or extreme-tail work. This distinction is invisible in the typical shell theorem alone.

Theorem 4 (Large deviations of history revelation)**.** *For fixed t and $p < 1/2$, $n^{-1}\log_2 R_{\text{hist}}$ obeys the contraction of the binomial type large-deviation principle under*

$$g(q) = \begin{cases} h_2(q), & 0 \le q \le 1/2, \\ 1, & 1/2 \le q \le 1. \end{cases}$$

In particular, for $\gamma \in [h_2(p), 1)$, let $q_\gamma \in [p, 1/2)$ solve $h_2(q_\gamma) = \gamma$. Then

$$\lim_{n\to\infty} -\frac{1}{n}\log_2 \Pr\left\{\frac{1}{n}\log_2 R_{\text{hist}} \ge \gamma\right\} = D_2(q_\gamma\|p). \tag{7}$$

At $\gamma = 1$ the rate is $D_2(1/2\|p)$. The same formulas give upper-tail bounds for Q_{FIBER} after the $O(\log n)$ shell shift.

Proof. The empirical error fraction E/m obeys the binomial large-deviation principle with rate $D_2(q\|p)$. The rank sandwich gives exponential equivalence to $g(E/m)$. The contraction principle yields the result; for an upper-tail threshold below one, the minimizing type is the unique $q_\gamma \ge p$ with $h_2(q_\gamma) = \gamma$. □

Corollary 2 (Fixed and exponentially rare quantiles)**.** *Every fixed quantile of $n^{-1}\log_2 R_{\text{hist}}$ converges to $h_2(p)$. A quantile whose upper-tail probability scales as $2^{-n\beta}$ has exponent $h_2(q_\beta)$, where $q_\beta \in [p, 1/2]$ solves $D_2(q_\beta\|p) = \beta$, until saturation at one.*

6 A typical-versus-mean complexity phase diagram

Suppose an independent exact code-side decoder has work bounded by

$$Q_{\text{code}} \le \text{poly}(n)2^{n(1-R)}.$$

For a high-rate linear code this is representative of a redundancy-state or syndrome-trellis search with 2^{n-k} states, although actual constants and code structure matter. Run FIBER and the code-side decoder in parallel and stop when either returns an exact certificate.

Theorem 5 (Hybrid complexity upper envelope)**.** *For fixed t and $p < 1/2$, the hybrid exact decoder obeys*

$$\limsup_{n\to\infty} \frac{1}{n}\log_2 Q_{\text{hybrid}}^{(\textit{fixed quantile})} \le \min\{h_2(p), 1 - R\}, \tag{8}$$

$$\limsup_{n\to\infty} \frac{1}{n}\log_2 \mathbb{E}[Q_{\text{hybrid}}^{\rho}] \le \min\{\rho H_{1/(1+\rho)}(p), \rho(1 - R)\}. \tag{9}$$

Proof. The hybrid work is no larger than either constituent work. Apply corollary 1 and theorem 3 to FIBER and the assumed code-side bound to the other constituent. □

The predicted fixed-quantile crossover is

$$h_2(p_{\mathrm{typ}}) = 1 - R,$$

while the predicted mean-work crossover is

$$H_{1/2}(p_{\mathrm{mean}}) = 1 - R.$$

Because $H_{1/2}(p) > h_2(p)$ for $p \in (0, 1/2)$,

$$p_{\mathrm{mean}} < p_{\mathrm{typ}}.$$

Table 2: Theoretical one-deletion/BSC operating thresholds. These are exponent boundaries, not finite-length guarantees.

Rate R	Redundancy $1-R$	$p_{\mathrm{typ}} : h_2(p) = 1-R$	$p_{\mathrm{mean}} : H_{1/2}(p) = 1-R$
0.625	0.375	0.07245	0.02254
0.750	0.250	0.04169	0.00903
0.875	0.125	0.01713	0.00205

This phase diagram predicts a real phenomenon observed in the reference computation: at $R = 0.75$, very low p produces strong median wins, while $p = 0.02$ or 0.04 can show substantial variability and worse mean/tail behavior even though $p = 0.04$ remains just below the typical asymptotic threshold.

7 Parameter complexity and the tractability boundary

The inverse history count contains

$$2^t \binom{n}{t} = \Theta(2^t n^t / t!)$$

for fixed t. The present decoder is therefore in the parameterized class XP: polynomial in n for each fixed t, with polynomial degree depending on t. It is *not* fixed-parameter tractable (FPT) in the formal sense $f(t)n^c$ with c independent of t.

If t grows linearly with n, then $n^{-1} \log_2 \binom{n}{t}$ contributes a nonzero deletion-position entropy and the fixed-edit theorem no longer preserves the BSC exponent. Any future FPT or positive edit-rate claim requires a new merging or dynamic-programming theorem. The general most-probable-string hardness literature makes an unrestricted exact claim implausible [11, 12].

8 Computational final gate

8.1 Exact and codebook-free implementations

The package contains three exact one-deletion decoders:

1. membership-only FIBER history enumeration;
2. a code-aware syndrome-trellis aggregate A* search;
3. a code-aware aggregate prefix A* search.

Random-linear and named CRC-polynomial codes use syndrome membership without a full codebook table. The faster of the two code-specific exact searches is the per-trial reference. Decoder order is randomized to reduce timing-order bias. Micro-worlds compare complete ML tie sets against exhaustive codeword likelihoods.

8.2 Reference-standard result

The package's frozen internal reference run produced

```
CONTINUE_HIGH_POTENTIAL_FIXED_EDIT_FIBER_TRACK_REAL_TRACE_REQUIRED
```

All exactness, moment/tail, phase, strong-baseline, and specialization-boundary gates passed. At the largest standard length $n = 40$, every primary trial returned the same complete ML tie set across the three exact decoders. Representative median FIBER-to-best-baseline wall-time ratios were:

Table 3: Reference-standard $n = 40$ mechanism probe. Only three trials were used per row; these are not publication-grade tail estimates.

Family	R	p	Median ratio	Observed 95% ratio	Interpretation
Random linear	0.75	0.005	0.157	0.176	strong low-p signal
Random linear	0.75	0.010	0.088	0.331	strong median; mean theory unfavorable
Random linear	0.75	0.020	0.207	0.340	positive finite sample, tail caution
Random linear	0.75	0.040	1.962	11.703	heavy-tail failure signal
Named CRC	0.75	0.005	0.267	0.288	strong low-p signal
Named CRC	0.75	0.020	1.155	1.363	baseline competitive
Random linear	0.875	0.005	0.093	0.179	strong median signal
Named CRC	0.875	0.010	0.222	0.439	positive median signal

The result supports a narrow low-substitution operating region. It does not establish $p99$ latency, compiled-code speedup, or asymptotic superiority. The standard profile is an early investment gate, not a paper-grade performance campaign.

8.3 Moment and tail numerical audit

The moment gate evaluates the exact finite-n rank sandwich through $n = 2048$, for $t \in \{1, 2\}$, multiple p, and $\rho \in \{1/2, 1, 2\}$. The reference run found:

- maximum finite-n reveal-moment exponent gap at $n = 2048$: 0.01648 bit/use;
- maximum certificate-upper exponent gap: 0.02276 bit/use;
- maximum fixed-quantile gap: 0.07211 bit/use;
- variational identity error below 5×10^{-16};
- decreasing finite-length gaps in every tested parameter sequence.

These computations verify the implementation and finite-length convergence; the proofs, not the plots, establish the theorem.

9 Honest specialization boundary: VT decoding

A generic code-modular decoder should not be expected to beat a code designed specifically for the impairment. The package therefore implements a classical linear-time single-deletion VT decoder, rather than the earlier $2n$ insertion-and-checksum enumerator. The exactness audit exhaustively verifies it through $n = 13$.

In the reference run, FIBER was approximately 22×, 43×, 86×, and 181× slower than the specialized VT decoder at $n = 15, 31, 63, 127$, respectively. This is a successful *boundary* result:

FIBER's value is code modularity and noisy fixed-edit ML, not replacement of an optimal specialized VT decoder.

VT codes are known to admit linear-time single-deletion decoding [16]. Any manuscript must report this result prominently rather than describe the VT gate as a performance win.

10 Physical plausibility and the unresolved real-trace gate

The package includes a physically motivated synthetic front end:

BPSK symbols → one accumulated timing-cycle slip → phase-dependent ISI → AWGN hard decisions.

A uniformly random initial phase and one-symbol accumulated drift create exactly one missed decision; ISI and AWGN create survivor substitutions. The standard held-out audit at $n = 48$, SNR 7 dB, and ISI strength 0.06 found:

- exactly one deletion in every frame;
- fitted substitution probability 0.001319 and held-out probability 0.001135;
- absolute fit error 1.84×10^{-4};
- held-out lag-one error correlation approximately -0.0012;
- deletion-position total variation 0.193 between train and held-out empirical laws.

This shows that a simple timing-slip front end can land inside the low-p theoretical region. It is not measured deployment evidence. The package deliberately contains no fabricated real trace and therefore cannot remove the qualifier

`REAL_TRACE_REQUIRED`

A measured trace file can be placed in `data/real_traces/`; the required CSV columns are `n`, `deleted_position`, and `substitution_weight`. Passing syntax and calibration checks is still only a prerequisite; end-to-end superiority over a matched receiver remains a separate experiment.

11 Final decision contract

11.1 Immediate stop conditions

The programme stops immediately if any of the following occurs:

1. one false exact certificate or complete-tie disagreement;
2. independent proof review invalidates the shell, moment, or tail theorem;
3. a primary source or patent contains the complete narrow theorem-and-algorithm synthesis and no stronger residual theorem remains;
4. equally optimized exact code-specific decoders dominate FIBER throughout the theorem-predicted favorable region in both median and high-tail latency;
5. no measured post-front-end impairment both fits the fixed-edit model and occupies a useful low-p operating region.

11.2 Narrow-paper outcome

If the theory survives but compiled performance or physical relevance fails, the authorized result is a fixed-edit foundations paper containing:

- the exact membership-first certificate;
- the deterministic shell theorem;
- the revelation moment and tail laws;
- the phase diagram and specialization boundaries;
- negative results outside the favorable region.

The multi-year field-defining claim must then stop.

11.3 High-potential continuation outcome

A high-potential continuation requires all internal gates to pass and authorizes only:

1. independent primary-source novelty adjudication;
2. independent mathematical proof review;
3. paper-grade compiled benchmarking with enough blocks to support $p99$ claims;
4. one measured post-front-end timing-slip or variable-length-demapping trace campaign.

It does not authorize general weighted transducers, hardware, or standards claims.

12 Reproducible package protocol

The supplied launcher performs the following operations:

1. locate the release ZIP in the Windows Downloads directory by SHA-256;
2. reject unsafe paths and symbolic links;
3. verify every internal manifest entry;
4. install only under `/home/afazeli2006/atom_coding/experiments/`;
5. create an isolated virtual environment outside the committed experiment folder;
6. run all unit tests and the frozen standard gate;
7. write a machine-readable verdict and exactly one command file;
8. commit and push only the experiment directory to the existing Git repository.

Installation, authentication, or Git failures are classified as

`EXECUTION_BLOCKED_NOT_A_SCIENTIFIC_VERDICT`.

13 Final scientific interpretation

The theoretical and computational evidence supports the following narrow statement:

> For a fixed number of uniformly located deletions and sufficiently low survivor-substitution uncertainty, codebook-free membership-only aggregate inverse search admits exact ML certificates and has a controlled probability-ordered history complexity. Its fixed-quantile exponent is governed by binary Shannon entropy, its history-revelation moments by binary Rényi entropy, and these yield distinct typical and mean operating boundaries relative to redundancy-state exact search.

This is a coherent new theorem-and-algorithm package with credible research potential. It is not yet a field-defining result. The decisive remaining uncertainty is whether a measured receiver impairment occupies the favorable region and whether the advantage survives equally optimized implementations and external novelty/proof review.

Reference investment command

```
CONTINUE_HIGH_POTENTIAL_FIXED_EDIT_FIBER_TRACK_REAL_TRACE_REQUIRED
FIELD_DEFINING_STATUS=HIGH_POTENTIAL_NOT_ESTABLISHED
NEXT_REQUIRED=external_proof_review,external_novelty_review,
              compiled_paper_grade_benchmark,measured_post_front_end_trace
STOP_IF_NEXT_GATE_FAILS=YES
```

A Proof-review checklist

An external reviewer should verify all of the following independently:

1. each unseen candidate has at most one compatible history per deletion set;
2. the sum-of-frontiers bound does not double count in a way that invalidates an upper bound;
3. membership-first scoring does not omit a competing codeword;
4. the strict certificate returns the complete tie set under the stated tolerance model;
5. 2^t is a history-count factor but not a channel-probability factor;
6. the shell shift uses $\lfloor \log_{(1-p)/p} \binom{n}{t} \rfloor$ and yields a strict inequality;
7. the history-rank sandwich is independent of within-shell tie order;
8. the moment variational identity is evaluated at $q_\rho < 1/2$;
9. the certified decoder obtains only a universal upper moment/tail bound, not equality;
10. the phase diagram is an upper-envelope theorem conditional on a $2^{n(1-R)}\,\mathrm{poly}(n)$ code-side decoder;
11. the current parameter dependence is XP, not FPT;
12. no field-defining or practical claim is inferred from the synthetic physical model.

B Primary-source novelty questions

The external novelty reviewer should answer claim by claim:

1. Does any paper or patent already give the same membership-only aggregate inverse generator and strict sum-of-stream-frontiers certificate?
2. Is theorem 2 already an explicit theorem in synchronization sequential decoding or fixed-deletion decoding?
3. Is theorem 3 a documented fixed-edit history-revelation theorem, or only derivable from generic guesswork after this representation is introduced?
4. Has the typical-versus-mean crossover for bounded-edit aggregate inverse decoding already been stated?
5. Does the proposed decoder add anything beyond direct implementation of general DMC candidate guessing [2]?
6. Are there patent claims broad enough to anticipate code-membership edit-history enumeration with likelihood aggregation and certified stopping?

C Reference-standard limitations

The internal standard run is intentionally inexpensive. It is suitable for an investment gate, not a paper:

- only three blocks were used at the largest primary length;
- Python wall time can favor one data structure over another;
- the strong exact baselines are independently organized but share channel and code utilities;
- $p95$ from three observations is not a statistically supported tail estimate;
- no measured front-end trace is bundled;
- no energy, cache traffic, or fixed-point implementation is measured.

A positive run therefore establishes only high potential and the right to perform the next external/physical gate.

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